

Design and Implementation of a Fully Automated ESP32-Based Tea Brewer

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github.com/tadeastehan/Automated-Tea-Brewer

Abstract

For students and busy users who need a large amount of tea ready for the whole day, existing tea makers either require constant manual attention or only prepare a single cup at a time. This paper presents the design and implementation of a fully automated tea brewing system capable of preparing 1.5 liters of tea with minimal user intervention. Unlike existing tea brewing solutions that focus on single-cup applications or require manual tea bag removal, this project addresses the practical need for large-volume, long-duration tea preparation. The system automatically heats water using induction cooking, submerges the mesh infuser with tea bag for optimal infusion, and automatically removes and disposes of the spent tea bag into a waste bin. An ESP32-based microcontroller system controls the entire process with a touchscreen interface and scheduling capabilities, allowing users to program tea preparation for specific times. The primary innovation lies in the compact mechanical design of the automated tea bag removal mechanism, utilizing a stepper motor with a threaded rod and dual-gear system. This solution overcomes critical engineering challenges including non-inductive glass compatibility, temperature measurement calibration, and mechanical precision in gear engagement. The completed prototype successfully demonstrates all core functionalities with reliable operation across multiple tea brewing cycles.

Solution and Implementation

The automated tea brewer integrates mechanical, electrical, and software components to create a seamless user experience. The system architecture comprises several interconnected subsystems: (1) an induction heating module that heats water in a standard glass teapot, (2) an infuser positioning mechanism for submerging and infusing tea bags, (3) temperature monitoring via infrared sensors, and (4) an ESP32-based control system with touchscreen interface and Wi-Fi connectivity.

Water Heating via Induction

Heating water efficiently and safely presented the initial design challenge. While submersible heating coils and

electric resistance elements were considered, they proved either operationally complex or difficult to control precisely due to slow thermal response times, which would result in excessively long brewing cycles. Induction cooking was selected for its rapid heating and instantaneous on-off control. However, standard glass teapots are non-conductive and incompatible with induction cooktops. The solution involved deploying a submersible stainless steel conversion adapter placed directly in the water. This approach maximizes heat transfer efficiency compared to contact-based adapters while requiring periodic cleaning. The induction cooktop is controlled via optoisolated phototransistor circuits that simulate user button presses, ensuring galvanic isolation between the high-voltage induction unit and the low-voltage control electronics.

The Automated Tea Bag Removal Mechanism: Core Innovation

The most significant technical achievement is the design of the automated tea bag removal and disposal system. Initial design iterations explored using a servo motor combined with a chain-drive mechanism to lift the mesh infuser, but this approach generated excessive mechanical stress on the chain and proved unreliable. A robotic arm configuration was subsequently evaluated but deemed excessively complex for the application. The final solution employs a stepper motor driving a threaded rod mechanism coupled with a dual-gear system to rotate the tea infuser linearly along the z-axis. This arrangement provides smooth, controlled vertical motion while minimizing mechanical complexity.

The gear design required meticulous engineering to achieve proper contact geometry and force distribution. Critical parameters included gear spacing calibration, threaded rod tightness adjustment, and bearing alignment. A linear bearing guides the z-axis carriage, ensuring smooth operation without binding or excessive play. Multiple iterations of 3D-printed prototypes were necessary to fine-tune the gear mesh tolerances and ensure reliable engagement between the circular gear and linear rack at the ejection position.

The stepper motor provides precise positional control, enabling the infuser to descend fully into the hot water, remain submerged for the programmed infusion duration,

and rise completely to position the tea bag over a waste bin. In the upper position, the circular gear wheel on the infuser engages with a fixed linear gear rack, which rotates the infuser and releases the used tea bag into the collection container. This single-motor, dual-gear approach elegantly solves the automation challenge while maintaining mechanical simplicity and high reliability compared to more complex chain-driven systems.

Temperature Monitoring and Calibration

Accurate temperature measurement is essential for scheduling delayed brewing and ensuring optimal tea infusion temperatures. The system implements an infrared (IR) thermometer pointed at the glass teapot surface. However, IR measurements detect surface temperature rather than internal water temperature. To establish the relationship between glass surface temperature and actual water temperature, a polynomial fitting procedure was performed: water temperature was measured independently while simultaneously recording IR surface readings across the full heating and cooling curve. A second-degree polynomial function was fitted to this data, enabling the firmware to calculate true water temperature from IR readings with an RMS error of 0.78°C. This calibration procedure accounts for the glass thermal properties and measurement angle effects.

Additionally, detailed heat dissipation measurements were conducted by recording temperature decrease over extended periods (10+ hours) to characterize how quickly water cools inside the teapot. These measurements are critical for the scheduler functionality: they enable the system to calculate precisely when to initiate heating so that the water reaches the target infusion temperature at the user's desired completion time, accounting for natural cooling during the brewing process.

Control System and User Interface

The control system employs two coordinated ESP32 microcontrollers. The primary ESP32 manages the touchscreen display and user interface, while the secondary controller ESP32 handles the critical automation tasks: driving the stepper motor via a dedicated stepper motor driver, controlling the induction cooktop through optoisolated circuits, acquiring infrared temperature measurements, monitoring pot presence via laser distance sensor, and maintaining timing through an integrated real-time clock (RTC) module. This dual-microcontroller architecture was necessary because the touchscreen display alone consumes a significant number of GPIO pins, leaving insufficient pins on a single ESP32 for all system peripherals and control functions.

The firmware implements a state-machine architecture managing the complete tea preparation sequence: heating initialization, temperature monitoring, infuser descent,

infusion timer, infuser ascent, and waste disposal. The user interface provides selection from six predefined tea profiles: green, black, herbal, fruit, functional, and white. Each profile includes customizable parameters for target infusion temperature and infusion duration. The scheduler feature enables users to set a desired completion time; the controller ESP32's RTC automatically calculates the required heating start time based on measured heat dissipation rates and initiates the process accordingly. Safety watchdog logic prevents heating without a detected teapot, enforced via the laser distance sensor. Multiple fail-safe mechanisms disable heating: absence of pot detection, sensor fault conditions, and excessive brewing duration beyond a predefined timeout. This layered approach ensures safe operation under all failure scenarios.

Conclusions and Future Work

This project successfully demonstrates that practical large-volume tea brewing automation is technically feasible and operationally superior to existing solutions. The completed system meets all primary objectives: reliable 1.5-liter brewing capacity, consistent quality across multiple brewing cycles, intuitive user control, and scheduled preparation capability. The most valuable contribution is the innovative tea bag removal mechanism, which provides a robust alternative to traditional chain-driven approaches and could serve as a reference design for other dispensing applications.

Significant challenges were encountered and resolved during implementation, including component sourcing delays (defective induction cooktop requiring replacement), thermal calibration complexity, and mechanical precision requirements for gear engagement. The choice of ASA material for 3D-printed components proved appropriate given thermal exposure and UV resistance requirements.

The primary incompleteness is the absence of a rotary carousel storage system for multiple tea bag varieties. Initial design iterations proved complex, and time constraints necessitated manual loading of individual tea bags into the infuser. This functionality remains clearly defined for future enhancement and represents the natural next step in the system's evolution.

Despite these limitations, the prototype delivers a genuinely unique solution addressing a practical gap in the market. The system proves that full automation at the 1.5-liter scale is achievable, differentiating it from all identified prior projects. The elegant mechanical design, robust embedded systems implementation, and user-centric interface collectively demonstrate successful engineering across multiple domains.